

NAME: _____ ID: _____

CSC 343 H5F – SOLUTIONS

Day Class 01

Instructor: Michael Liut

DURATION: 2 hours

University of Toronto

Midterm Examination

October 31st, 2018

Read all instructions before starting the exam

This test paper includes 15 pages and a total of 18 questions; 13 true-or-false questions, 1 fill-in-the-blank question, and 4 questions requiring a written answer. You are responsible for ensuring that your copy of the paper is complete. Bring any discrepancy to the attention of your invigilator.

Special Instructions :

1. The answers to the written questions (Questions 15-18) must be answered in the space provided (i.e. immediately following the question). Written answers are to be clear and concise; illegible solutions will not be marked.
2. Read all questions before starting the exam – some are easier than others.
3. The True-or-False section requires you to illustrate the validity by an example, or disprove the validity by a counter-example. If your justification is proof by definition, this is perfectly valid!
4. **This is a closed book exam.** No memory aids, notes, calculators, or, textbooks of any kind are allowed during the test. The use of any electronic device is strictly prohibited. Failure to comply will result in your immediate dismissal from the examination and a grade of 0.
5. Students are not allowed to be involved in any communication of any kind. Questions concerning this exam must be brought to the attention of the invigilator(s) or the instructor. Any attempt of alternative communication will be considered a case of academic dishonesty.
6. No unauthorized scrap paper is allowed to be used. The invigilator(s) will supply every student with needed scrap paper when asked. An additional mark will be awarded to students who circle this line; do not look around, just do it.
7. Documents to be returned: this questionnaire and all scrap paper if used. All of these documents must bear the student's name and number. Only the face page of the questionnaire needs to bear the student's name and number.

question(s)	mark	out of
1 – 14		20
15		10
16		10
17		10
18		3
misc		n/a
total		53

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Question 1 – 13 are True-or-False questions. You are required to illustrate the validity by an example, or disprove the validity by a counter-example.

For each question only select one answer (i.e. True or False). To select an answer, you must circle the word “True” or the word “False”. The negative marking scheme is not used. Incorrect or missing answers earn a mark of 0. Therefore, do not leave any questions blank. For full marks you must justify (prove/disprove) your answer.

Question 1 [1.5 marks]

A weak-relation $W(ForeignKey, Attribute_2, \dots, Attribute_n)$, with n -attributes, has a *ForeignKey* which references the *PrimaryKey* from strong-relation $S(PrimaryKey, Attribute_2, \dots, Attribute_n)$, with n -attributes. This means, minimally, that one of W 's attributes must always also be part of the key for W (i.e. W 's key would be: $\{ForeignKey, Attribute_i\}$, where $Attribute_i$ is an arbitrary attribute belonging to W).

A. True.

$\Rightarrow$ B. False.

FALSE. W can utilize $S(PrimaryKey)$ as the sole key, without the need for its own attribute(s), as its minimal key.

Marking Scheme:

[0.5 marks for identifying this statement is FALSE]

[1 mark for justifying that the statement is FALSE]

[mark of 0 if student identifies the answer as being TRUE]

Question 2 [1.5 marks]

A relation is made up of two components: (1) an instance (i.e. a table defined by its cardinality and degree/arity); and (2) a schema (i.e. the relation name and the type of each column).

$\Rightarrow$ A. True.

B. False.

TRUE. Proof by definition.

Marking Scheme:

[0.5 marks for identifying this statement is TRUE]

[1 mark for justifying that the statement is TRUE]

[mark of 0 if student identifies the answer as being FALSE]

Question 3 [1.5 marks]

Physical data independence separates the physical (refers to the hardware-level/system design) and conceptual schemas. Logical data independence separates the conceptual schema from the external schema, which depends on relations in the conceptual schema.

- $\Rightarrow$ A. True.
B. False.

TRUE. Proof by definition.

Marking Scheme:

- [0.5 marks for identifying this statement is TRUE]
- [1 mark for justifying that the statement is TRUE]
- [mark of 0 if student identifies the answer as being FALSE]

Question 4 [1.5 marks]

Given two relations R and S , the Cartesian Product is denoted $R \times S$, where R and S are sets: $R \times S = \{(r, s) : (r \in R) \text{ and } (s \in S)\}$. Thus, in general, $(R \times S) = (S \times R)$ (i.e. the Cartesian Product operation is commutative).

- A. True.
 $\Rightarrow$ B. False.

Answer:

$$R \times S = \{(r, s) : (r \in R) \text{ and } (s \in S)\}$$

Note: The Cartesian Product of two sets is a set, and the elements of that set are ordered pairs. In each ordered pair, the first component is an element of $\mathbf{R}$, and the second component is an element of $\mathbf{S}$.

For example, let:

$$\begin{aligned} R &= 1, 2, 3 \text{ and } S = (w, x), (y, z), \\ R \times S &= (1, 2, (w, x)), (1, 2, (y, z)), (3, (w, x)), (3, (y, z)). \end{aligned}$$

If $\mathbf{R}$ and $\mathbf{S}$ are both finite sets, then $|R \times S| = |R| \cdot |S|$ as there are $|R|$ choices for the first component of each ordered pair and also $|S|$ choices for the second component of the ordered pair. Therefore, Cartesian Product is **not commutative** for the sets $\mathbf{R}$ and $\mathbf{S}$. This is justified in the example above proving that $(R \times S) \neq (S \times R)$.

FALSE.

Marking Scheme:

[0.5 marks for identifying this statement is FALSE]

[1 mark for justifying, proof above, that the statement is FALSE]

[mark of 0 if student identifies the answer as being TRUE]

Question 5 [1.5 marks]

A domain constraint expresses conditions on the values of each tuple, independent of other tuples. A tuple constraint is a domain constraint that involves a single attribute.

A. True.

$\implies$ B. False.

Answer:

FALSE. The definitions for domain constraint and tuple constraint are reversed.

Marking Scheme:

[0.5 marks for identifying this statement is FALSE]

[1 mark for justifying that the statement is FALSE]

[mark of 0 if student identifies the answer as being TRUE]

Question 6 [1.5 marks]

Triggers are more commonly used to maintain database consistency, alert us to unusual events, allow us to impose flexible constraints, and can even generate a log of events to support auditing and security checks. Where as *Constraints* prevent inconsistency by any kind of statement, afford the DBMS the opportunity to optimize, and are easier to understand than Triggers.

$\implies$ A. True.

B. False.

TRUE. By definition.

Marking Scheme:

[0.5 marks for identifying this statement is TRUE]

[1 mark for justifying that the statement is TRUE]

[mark of 0 if student identifies the answer as being FALSE]

Question 7 [1.5 marks]

With respect to *Bag Law*, it can be said that the Idempotent Law¹, in general, for union does hold for bags (i.e. given some relation R , then $R \cup R = R$).

A. True.

$\implies$ B. False.

FALSE. By counter-example: $\{1\} \cup \{1\} = \{1, 1\} \neq \{1\}$

Marking Scheme:

[0.5 marks for identifying this statement is FALSE]

[1 mark for justifying that the statement is FALSE]

[mark of 0 if student identifies the answer as being TRUE]

Question 8 [1.5 marks]

Suppose you have two relations R and S , where a tuple of R that has no tuple of S with which it joins is said to be dangling. Assuming that we take R and JOIN it to S (i.e. $R \bowtie S$), then to preserve² each tuple in R we can perform any of the following JOINS (independently): “OUTER”, “RIGHT”, “INNER”, or “CROSS”³.

A. True.

$\implies$ B. False.

FALSE. JOINS: “OUTER” or “LEFT”

Marking Scheme:

[0.5 marks for identifying this statement is FALSE]

[1 mark for justifying that the statement is FALSE]

[0.5 marks in penalty for each incorrect item on the list]

[mark of 0 if student identifies the answer as being TRUE]

¹real definition: an element x of some operator $\cdot$ is said to be idempotent if $x \cdot x = x$. If all elements are idempotent with respect to $\cdot$, then $\cdot$ is called idempotent. i.e. $\forall x, x \cdot x = x$ then idempotent law holds for $\cdot$.

²your join may pad values with NULL, but cannot lose any information from R 's tuples

³real definition: CROSS JOIN is equivalent to taking the Cross Product

Question 9 [1.5 marks]

Given two of relations **R** and **S**, in the context of data modeling, there are only three degrees of cardinality that can be between them: (1) one-to-one ($1 : 1$); (2) one-to-many ($1 : M$); and (3) many-to-many ($M : N$).

A. True.

$\Rightarrow$ B. False.

FALSE.

Marking Scheme:

[0.5 marks for identifying this statement is FALSE]

[1 mark for justifying that the statement is FALSE]

[mark of 0 if student identifies the answer as being TRUE]

Question 10 [1.5 marks]

Let's assume that we are working with subset of a banking schema. Given two of the entities: **Customer** and **Loan**, and their relationship **Borrower**, it is fair to assume that there is a partial participation between **Customer** — **Borrower** and clear to assume a total participation between **Borrower** == **Loan**.

⁴

$\Rightarrow$ A. True.

B. False.

TRUE. Not every bank customer will have a loan, however, every Loan will have an associated bank customer (i.e. "borrower").

Marking Scheme:

[0.5 marks for identifying this statement is TRUE]

[1 mark for justifying that the statement is TRUE]

[mark of 0 if student identifies the answer as being FALSE]

⁴real definition: — indicates partial participation and == indicates total participation

Question 11 [1.5 marks]

The Database Administrator (DBA) is responsible for ensuring these critical tasks: (1) the security and authorization of the data on the DBMS; (2) the data's availability and recovery from any failure; and (3) database tuning, which will evolve over time. The DBA, however, is not responsible for the design of the conceptual and physical schemas.

A. True.

⇒ B. False.

FALSE. The DBA is also responsible for the design of the conceptual and physical schemas.

Marking Scheme:

[0.5 marks for identifying this statement is FALSE]

[1 mark for justifying that the statement is FALSE]

[mark of 0 if student identifies the answer as being TRUE]

Question 12 [1.5 marks]

Cardinality, with respect to uniqueness of data, says that something has low cardinality if it has a low number of duplicates in the data and high cardinality if the data has a high number of duplicates.

A. True.

⇒ B. False.

FALSE. If a column (of data) has a lot of duplicates, then it has a low cardinality, but if the values are highly unique (e.g. Social Insurance Number) it has a high cardinality.

Marking Scheme:

[0.5 marks for identifying this statement is FALSE]

[1 mark for justifying that the statement is FALSE]

[mark of 0 if student identifies the answer as being TRUE]

Question 13 [1.5 marks]

In an entity-relationship diagram, the ISA hierarchy is similar to the parent-child relationship in object oriented programming and is organized in a tree-like structure. Therefore, it is fair to assume that if a parent relation P has three children relations X, Y, and Z, then any member of P will not be members of more than one child (e.g. $(\exists a \mid (a \in P) \text{ and } (a \in X) \implies (a \notin Y) \text{ and } (a \notin Z)))$).

A. True.

$\implies$ B. False.

FALSE.

Marking Scheme:

[0.5 marks for identifying this statement is FALSE]

[1 mark for justifying that the statement is FALSE]

[mark of 0 if student identifies the answer as being TRUE]

Question 14 [0.5 marks]

Fill in the blank: the name⁵ of one of my Teaching Assistants is _____.

⁵first name, last name, and/or common name will be taken as correct

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Questions 15 – 18 are questions that require a written answer. The answers are to be written in the space provided below each question.

Question 15 [10 marks]

Consider the relations: **Students**, **Faculty**, **Courses**, **Rooms**, **Enrolled**, **Teaches**, and **MeetsIn**; more precisely defined as:

Students(sid: string, name: string, login: string, age: integer, gpa: real)

Faculty(fid: string, fname: string, sal: real)

Courses(cid: string, cname: string, credits: integer)

Rooms(rno: integer, address: string, capacity: integer)

Enrolled(sid: string, cid: string, grade: string)

Teaches(fid: string, cid: string)

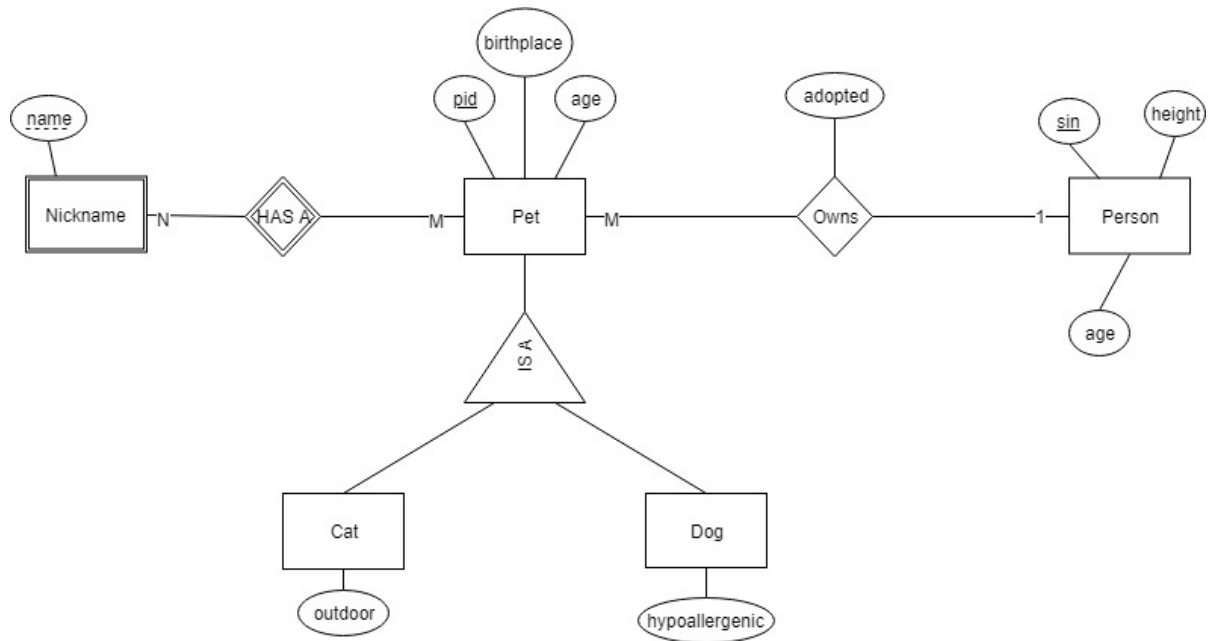
MeetsIn(cid: string, rno: integer, time: string)

- (a) [5 marks] Create an entity-relationship diagram that denotes the relations given above.
- (b) [3 marks] List all the foreign key constraints among these relations.
- (c) [2 marks] Give an example of two possible domain constraints on the relations above.

Question 16 [10 marks]

Given the following Entity-Relationship Diagram (ERD), in Chen's Notation below, you must translate the ERD to its appropriate Data Definition Language (DDL).

Requirements: (1) any attribute requiring a "Yes" or "No" answer should be reflected as a boolean value, all others must be reasonably determined (type and size); (2) you must enforce that *a Dog cannot be a Cat* and *a Cat cannot be a Dog*; (3) no intra-relational constraints need be created for any relation; (4) all inter-relational constraints must be enforced; and (5) each time an entry is removed/deleted all of its associated entries are also removed/deleted.



Answer:

See file: `mtqDDL Pets.ddl`.

Marking Scheme:

- [3 marks for creating all 6 tables with correct attributes of appropriate types/lengths]
- [1 mark for properly identifying PKs for each table]
- [1 mark for properly identifying FKs]
- [1 mark for correctly constraining Cat and Dog with Pet.pid]
- [2 marks for implementing proper cascading events]

Question 17 [10 marks]

Given the following DDL:

```

CREATE TABLE Table_1 (
    A int
    B int
    C int
    PRIMARY KEY(A)
);

CREATE TABLE Table_2 (
    D int
    E int
    F int
    PRIMARY KEY(D,F)
);

delimiter $$
CREATE TRIGGER trig AFTER INSERT ON Table_1
    FOR EACH ROW
    BEGIN
        INSERT INTO Table_2(F, E, D) VALUES(0, NEW.A, NEW.A + 5)
    END;
$$ delimiter;
```

For each question below, assume that the DML given is run on a clean version of the DDL (above); that is, each part does not affect the other. With all the information provided, your task is to provide the output⁶ of the following DML executions:

- (a) [2 marks] INSERT INTO Table_2 VALUES(1, 7, 9);
- (b) [3 marks] INSERT INTO Table_1 VALUES(1, 2, 3);
- (c) [3 marks] INSERT INTO Table_1(B,A) VALUES(4, 5);
INSERT INTO Table_1(A,C) VALUES(4, 5);
- (d) [2 marks] INSERT INTO Table_1(B,A) VALUES(4, 5);
INSERT INTO Table_1(B,A) VALUES(10, 9);
INSERT INTO Table_2(F,E,D) VALUES(0, 0, 24-10);
INSERT INTO Table_1(A,B) VALUES(10, 9);

⁶you may opt to draw the tables content or write out what each table contains. In the event of an error, you must indicate what that error is and what caused it.

Answer:

- (a) [2 marks] Error on Foreign Key constraint, 7 does not exist in `Table_1`.
- (b) [3 marks] `Table_1` has the values inserted as A:1 B:2 C:3
`Table_2` has the values inserted as D:6 E:1 F:0
- (c) [3 marks] `Table_1` has the values inserted as A:4 B:NULL C:5 and A:5 B:4 C:NULL
`Table_2` has the values inserted as D:9 E:4 F:0 and D:10 E:5 F:0
- (d) [2 marks] Error on Primary key, caused by the third line. PK (14,0) already exists.
BLA BLA.

Marking Scheme:

[marks goes to yada yada]

Question 18 [3 marks]

Consider the schema below about movies and actors (keys are underlined):

```
Actor(aid, name)
Movie(mid, title, year, genre, length)
Roles(aid, mid, character)
```

You are given an SQL query which finds all actors who did not star in any movie (i.e. play a character) in the year 2018 and only returns the actor's name and actor ID:

```
SELECT aid, name
FROM Actor
WHERE aid NOT IN
      (SELECT roles.aid
       FROM Actor, Roles, Movie
       WHERE year = 2018 AND actor.aid = roles.aid
          AND roles.mid = movie.mid);
```

Your task is to translate the SQL query (above) into Relational Algebra.

Answer:

Marking Scheme:

- [1 mark for identifying the correct operators]
- [1 mark for identifying the correct operands (i.e. relations/tables)]
- [1 mark for selecting the appropriate data]

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